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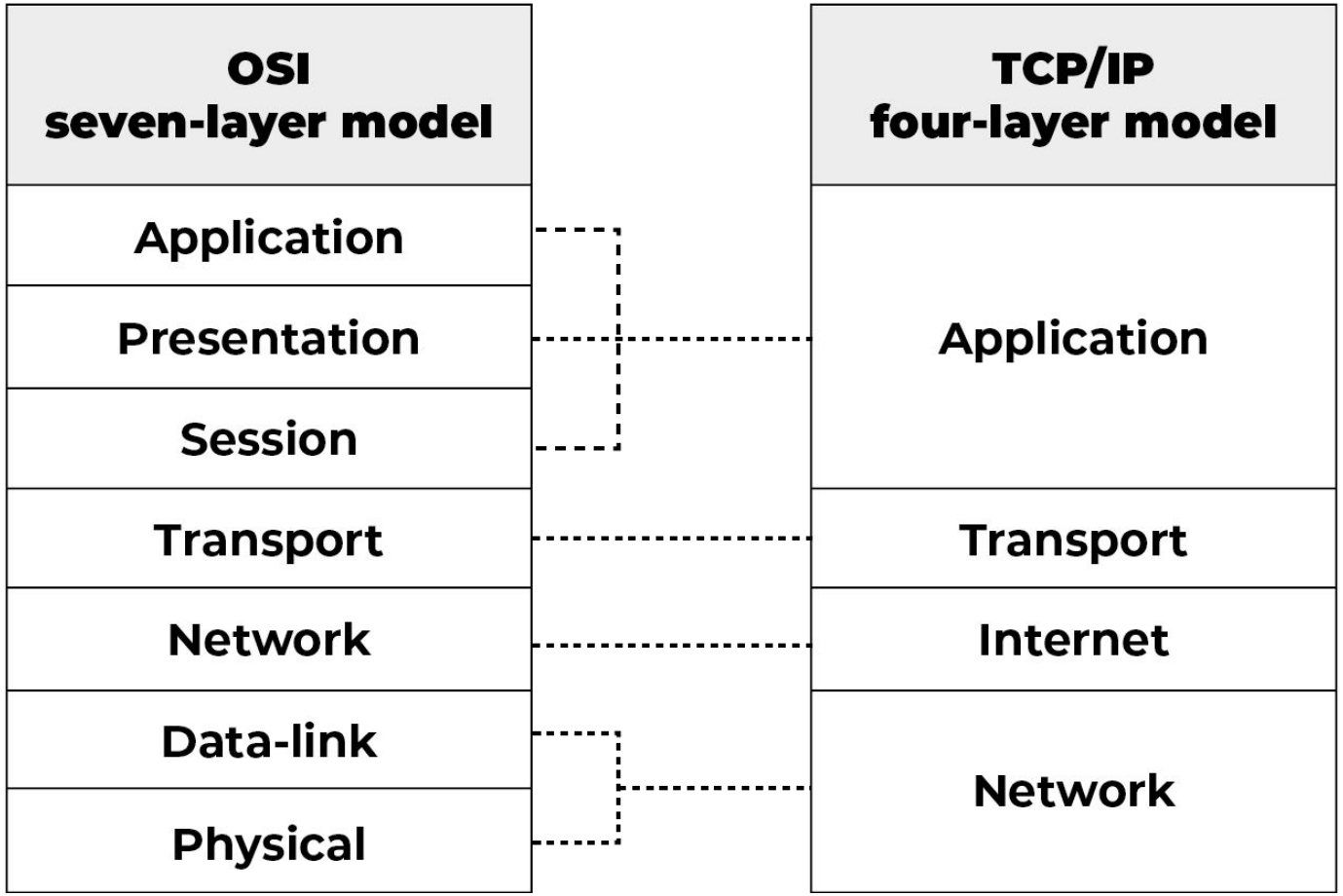
OSI vs TCP/IP models (from here)

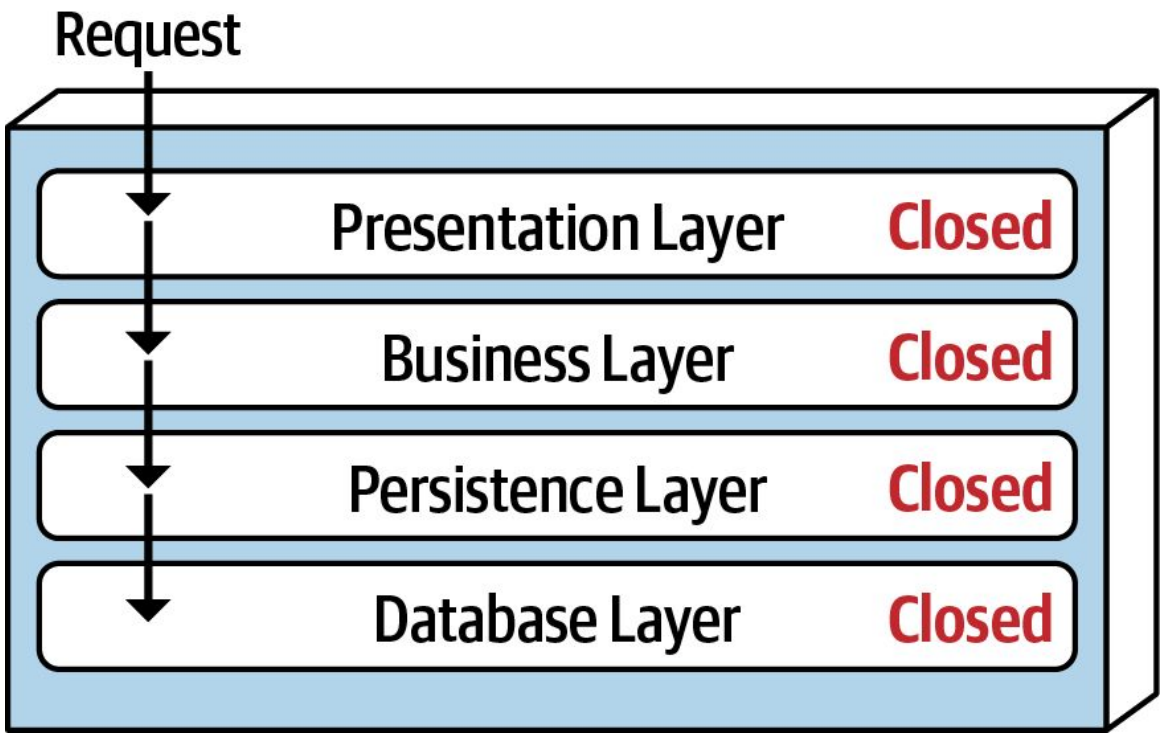


- Only adjacent layers can communicate directly
- Therefore, Each layer
  - Makes the overall system slower
  - (Should) simplify logical reasoning
- There is no best solution
  - Practice decides
- Exactly the same happens with n-tier architectures

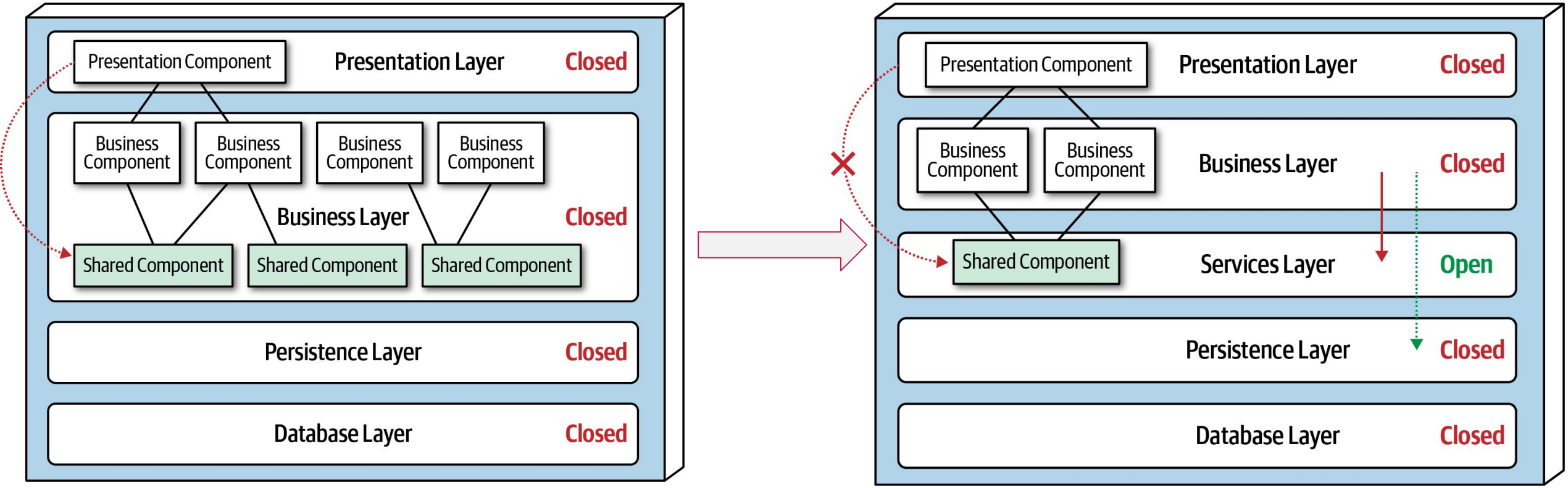
https://en.wikipedia.org/wiki/OSI_model https://en.wikipedia.org/wiki/Internet_protocol_suite

Isolating layers

- Layers are either open or closed
  - Closed: the upper layer can communicate only with the lower one
  - Open: the layer can be "skipped"
- Layers of isolation
  - Changes in one layer do not affect other layers if contracts do not change
  - Otherwise, affect only upper layer
    - If this is closed!
- Performances vs coupling
  - Each closed layer adds more steps
  - Each open layer allow layers to depend on more than one



Adding layers



- Adding layers usually require they are open
  - Otherwise they should either expose all the interfaces of the lower layer, or the upper layer should be mostly rewritten

Summary of the layered architecture

- Good starting point to develop the system
- Reasonable and very natural choice
  - Easy to assign who should what inside the organization
- Still widely used (maybe for specific distributed subcomponents)
- Often architecture sinkhole antipattern appears
  - Layers are just forwarding calls to the lower layers
    - Slowing down the execution without improving decoupling
- In general, good choice for small (Web) applications
- A minimal example:
  - https://www.baeldung.com/spring-boot-angular-web

Your wooclap poll will be displayed here

1

Install the **Chrome** or **Firefox extension**

2

Make sure you are in **presentation mode**

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https://app.wooclap.com/events/SADM/0 question 13

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Layered (n-tier) architecture rating

1	Overall cost	3.1
2	Simplicity (structural)	3.6
3	Modularity ...	3.7
4	Maintainability ...	4.3
5	Testability ...	4.1
6	Deployability ...	3.2
7	Evolvability ...	3.5
8	Responsiveness ...	3.2
9	Scalability ...	3.7
10	Elasticity ...	3.1
11	Fault tolerance ...	3.6